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AgroSat: a precision farming tool for managing in-field variation of optimal input rates and supporting agronomic decisions

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ABSTRACT

AgroSat is a free web-based platform aimed at providing users with a precise data-driven tool for managing key agricultural operations. This decision support system (DSS), mainly dedicated to arable crops, has a user-friendly interface designed to facilitate the adoption of digital agriculture by researchers and end-user farmers. The platform uses publicly available and continuously updated satellite imagery (Sentinel-2 and MODIS), global gridded climatic data (Copernicus Climate Data Store), regional agrometeorological data repository (OpenData), and georeferenced data to provide cost-efficient operational services at field level and support research-oriented applications. Integrating multi-source data, AgroSat allows monitoring crop phenology, water status, vegetation index, and weather forecasts. Moreover, it further supports decision-making through generating downloadable output spatial products, such as yield potential and input prescription maps. The present research aims to introduce the products and services of AgroSat platform, highlighting its application for nitrogen (N) fertilization based on variable rate technology (VRT) through a case study on winter wheat. Research results summarized the potential of AgroSat as an accurate input management tool for generating zonal N prescription maps and stabilizing within-field yield and crop vigor, supporting the overall crop performance and resource use efficiency. Future platform improvements aim to further support sustainable farming practices through additional services, such as soil organic carbon estimation and proposing new strategies for accurate input management.

1. Introduction

The optimization of agricultural inputs is fundamental for effective precision agriculture, which aims to address the long-term environmental impacts of conventional farming by supporting resource-efficient production systems. Precision agriculture is officially defined as a management strategy that takes account of temporal and spatial variability to improve sustainability of agricultural production [1]. As an integral part of precision agriculture, remote sensing plays a key role in revolutionizing sustainable management of agricultural lands [2]. This cutting-edge technology allows growers to collect, visualize, and assess crop, soil, and weather conditions at different stages of production in a cost-effective manner. Advanced remote sensing technologies offer various options of both platforms (airborne and spaceborne) and sensors (visible, multispectral, hyperspectral, and thermal) to collect large amounts of agricultural data with different spatial, temporal, and spectral resolutions [3]. For instance, unmanned aerial vehicles (UAVs) and satellites equipped with high-resolution sensors capture detailed imagery and provide precise positioning data to address specific agricultural needs and challenges [4]. Remote sensing applications include crop mapping and yield prediction [5,6], crop health [7], soil properties [8], weather forecasting [9], pest and diseases [10], and plant biodiversity [11].

Since 1970s, satellite remote sensing has become one of the major space-based technologies used for regional and global crop monitoring [12]. At present, various operational global scale agricultural monitoring systems provide open and interoperable services to support decision making focusing on crop production components [13–15]. Such systems include the Global Agriculture Monitoring (GLAM) Project [16], the Global Information and Early Warning System (GIEWS) [17], CropWatch [18], GEOGLAM [19], the Anomaly Hot Spots of Agricultural Production (ASAP) System [20], and most recently the Agro-informatics platform [2].

Notably, the agricultural sector has witnessed a digital

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transformation in recent years that has had evident impacts on social, economic, and environmental aspects. Agricultural digitalization applies to the integration process of emerging technologies into the farming system, such as the Global Positioning Systems (GPS), Geographic Information Systems (GIS), Internet of Things (IoT), artificial intelligence (AI), big data analytics, robotics, and communication networks [21]. The scientific literature has primarily focused on the technical aspects of applying these promising tools to optimize the agricultural input use, enhance the production operations, food systems, and value chains [22,23]. To operationalize these data-driven insights, farm management information systems (FMIS) provide digital platforms to integrate, organize, and visualize multi-source agricultural data [24]. Platforms such as decision support systems (DSS) function as analytical modules that transform integrated data into practical recommendations, to guide site-specific and strategic farm management decisions [25,26]. DSS is defined as a computer-based system that integrates and processes multi-source data, aiming at supporting farmers in their decision-making [27–29]. It is worth mentioning that such platforms must be user-friendly and adapted to the socio-technical system of farmers' needs and practices [30].

DSS may leverage AI and machine learning (ML) techniques to enhance data integration and prediction accuracy, enabling to build on past data and respond to new situations [31]. The process of integrating multi-source data is known as data fusion, which aims to produce more accurate and concise information than that provided by a single data source [32]. Research has demonstrated that combining remote sensing data with geospatial tools and ground truth information strengthens the accuracy of agricultural monitoring and enhances the spatio-temporal resolution of acquired data [2,23,33]. Data fusion approach has been used for various agricultural applications using data collected from proximal, aerial or orbital scales [34]. Such applications include crop monitoring [35], phenological assessment [36], soil characterization [37], yield prediction [38], disease detection [39], and crop evapotranspiration [40].

In recent years, several web-based DSS platforms have been developed to reach cost-effective, more accurate, and portable diagnosis systems. Among these, an agro-climatic DSS tool called SIMAGRI was developed to analyze agriculture-related climate risks and simulate crop production models in Uruguay [41]. CropGIS web-application was built for the spatial and temporal visualization of crop biomass development and predictions using historical meteorological data and crop growth model [42]. AgroDSS was created as a cloud-based DSS toolbox that allows farmers to upload and analyze their own data, offering several statistical tools for predictive modeling and time series analysis [43]. LANDSUPPORT was developed as a free, web-based geospatial DSS that promotes agricultural and environmental sustainability through the implementation of effective land use and management policies [44]. Recently, a detailed overview of the architecture proposed for a web-based spatial DSS for delineating management unit zones (MUZs) and crop monitoring has been developed [45].

Regarding the optimization of agricultural inputs, precision techniques such as the Variable Rate Technology (VRT) allow efficient input applications (e.g., water, fertilizers, pesticides) in response to the within-field variability [46]. VRT fertilization has been shown to increase nitrogen (N) use efficiency by improving crop yield, quality, and profitability while reducing fertilizer use [47]. Map-based variable rate N application based on data fusion of soil and crop parameters enabled site-specific management of fertilization in barley and wheat [48]. Recent DSS platforms that support crop nutrient management include CropSAT, a free web platform that generates satellite-based vegetation index maps for N fertilization in cereals, but its use is limited to Sweden and Denmark [49]. Another ontology-based DSS tool, designed for use in Bavaria, Germany, relies on regional agricultural data sources to automatically recommend an application time and rate for the N fertilization of winter wheat [50]. Similarly, a region-specific prototype DSS was tested and evaluated for N fertilization management of winter wheat in Germany [51].

While several existing DSS platforms support crop monitoring or nutrient management, most remain region-specific or rely on limited data streams and single-function analytical modules. AgroSat is designed as an integrated and transferable framework that combines multi-source data fusion of satellite observations, agro-meteorological records, and user-generated georeferenced field data within a unique web-based environment. Through automated spatial and temporal harmonization of heterogeneous datasets, the platform enables dynamic field-scale assessment of crop conditions and supports decision-making across diverse contexts. A distinctive feature of AgroSat is the implementation of two complementary fertilization management modes, compensation and differentiation, which allow users to evaluate and operationalize alternative variable-rate strategies based on intra-field variability and crop growth dynamics.

In response to the growing need for accessible, data-driven decision support tools, this study aims to introduce AgroSat to facilitate the adoption of digital agriculture by growers for managing key farming operations. The study outlines the functional services and products of AgroSat in detail, highlighting their applicability for monitoring crop growth dynamics and supporting informed decision-making in agricultural management. A case study was included to demonstrate the practical implementation of AgroSat and its potential in detecting intra-field variability in crop requirements, guiding variable-rate N fertilization strategies, generating zonal prescription maps, and stabilizing within-field yield potential, with the overall aim to improve crop performance and resource use efficiency.

2. AgroSat data, products, and system architecture

AgroSat is a free, web-based platform developed by the Italian National Research Council - Institute of BioEconomy (Digital Agriculture, AgroSat, IBE-CNR; https://www.agrosat.it/) to provide an open, precise agricultural tool for managing key agricultural operations [52]. It is mainly dedicated to monitoring arable crops, with the possibility to cover tree crops for research-oriented applications. Currently, AgroSat is available both in Italian and English versions, covering Italy nationwide, some regions of European (EU) and Extra-EU countries, including France, Germany, Austria, Slovakia, Czech Republic, Romania, Hungary, Croatia, New Zealand, Japan, and Korea. The platform has a user-friendly design with a level of complexity that can be adapted to users of varying expertise and needs. Applying digital technologies, AgroSat aims to increase accessibility to multi-temporal data and services, enabling farmers to make informed agronomic decisions and supporting accurate input management. This platform relies on the integration of multi-source data, including satellite imagery, geographic and land use data, agrometeorological observations, and seasonal forecasts to provide a comprehensive spatial monitoring of crop conditions and intra-field heterogeneity. Leveraging these integrated data sources, AgroSat permits the estimation of potential yield, monitoring phenological development and water balance, the prescription of fertilization maps, and many other operational services detailed in the following sections. Table 1 summarizes the key data and products available in AgroSat.

2.1. Data and products

2.1.1. Satellite data

AgroSat leverages publicly available satellite imagery to derive essential vegetation and environmental indicators for precision agriculture. The platform primarily integrates data from the Sentinel-2 constellation operated by the European Space Agency (ESA), composed of three polar-orbiting satellites, Sentinel-2A/B/C, launched in 2015, 2017, and 2024, respectively. All satellites carry the Multispectral Instrument (MSI), which acquires imagery across 13 spectral bands with spatial resolutions of 10, 20, and 60 m [53]. To expand the

Table 1
Summary of multi-source data and end-user products available on AgroSat.

Group	Data / Product	Source	References
Multi-source data			
Satellite data	Sentinel-2A/B/C (Bands B2, B3, B4, B5, B6, B7, B8, B8A, B11, B12)	Sentinel-2 ESA	[53]
	MODIS (Bands B1, B2)	MODIS - NASA	[54]
	Land surface parameters	LSA-SAF (EUMETSAT)	[55]
Geographic / Field data	Production Unit (PU)	User-defined in AgroSat	
	Field notes	User input in AgroSat	
Land use	CORINE Land Cover (CLC)	Copernicus CLMS (EEA)	[56]
	Land use and cadastral records	National/regional datasets	
Gridded weather data	CDS climate variables	Copernicus CDS-API	[57]
	ERA5 Reanalysis	Copernicus CDS-API, ECMWF	[57,58]
	SEAS5 Seasonal Forecasts	Copernicus CDS-API, ECMWF	[59]
Data cleaning	Cloud and shadow detection (AgroShadow tool)	Sentinel-2 ESA (SWIR, NDVI)	[60]
End-user products and services			
Maps	Natural color image	Sentinel-2 ESA (RGB)	[53]
	Surface soil moisture (SSM) estimates	OPTRAM, Sentinel-2 ESA (SWIR, NDVI)	[53,61, 62]
	Vegetation index (NDVI)	Sentinel-2 ESA, MODIS	[53,54]
	Productive potential map (%)	Sentinel-2 ESA (NDVI)	[53,63, 64]
	Variable-rate N fertilization map	Sentinel-2 ESA (NDVI), GPS-based yield data	[53, 65–67]
Meteorological data	Weather observations and forecasts	Agrometeorological networks, Copernicus CDS-API	
Crop phenology	Phenological growth	Agrometeorological networks, Crop models	[68]
Water balance	ET_0 and rainfall	LSA-SAF (EUMETSAT), FAO, Agrometeorological networks, Copernicus CDS	[55,68, 69]
Seasonal forecast-based yield prediction	Yield and quality prediction	SEAS5 Seasonal Forecasts, Crop models	[59,63]

temporal depth of monitoring, AgroSat also incorporates Moderate Resolution Imaging Spectroradiometer (MODIS) imagery from NASA. MODIS products provide broader coverage with a coarser spatial resolution (250–1000 m), 36 spectral bands ranging from 0.405 to 14.385 μm, and a high temporal resolution with near-daily observations [54]. These datasets complement Sentinel-2 imagery for long-term vegetation trend analysis. AgroSat uses the bands B2, B3, B4, B8, and B12 from Sentinel-2, and B1 and B2 from MODIS for the end-user products. In addition to reflectance-based indices, AgroSat employs land surface parameters derived from the Land Surface Analysis Satellite Applications Facility (LSA-SAF) of EUMETSAT [55]. Specifically, it uses reference evapotranspiration (ET_0) products generated by the SEVIRI sensor onboard the Meteosat Second Generation (MSG) satellites. These products cover Europe and Africa with a spatial resolution of 3–5 km and are distributed operationally every 30 min with a 1-day latency [55].

2.1.2. Geographic data

In addition to remote sensing, AgroSat incorporates proximal data sources to enhance field-specific monitoring and decision support. User-provided data represents a key input for AgroSat's customization and geolocation features. Upon registration, users can access a personal dashboard that allows the creation and storage of Production Units (PUs). Each PU corresponds to a single agricultural field, whose boundaries can be defined interactively via a drawing tool (Fig. 1, a) or by uploading georeferenced shapefiles (shp, dbf, shx, prj). To ensure consistent spatial processing, each PU must not exceed 200 ha. Users can also geolocate specific in-field observations (e.g., phenological stages or anomalies) using the field marker function. Additionally, registered users can create a field notebook dedicated to each PU, enabling access to all tools available through the AgroSat-Studio environment.

2.1.3. Land use maps and gridded data

Land use data plays a crucial role in the AgroSat platform for preprocessing and masking satellite information. Non-agricultural land classes and crop types not covered by AgroSat's modeling capabilities are excluded to ensure data consistency and avoid misclassification in downstream analyses. AgroSat includes both national/regional land use datasets (e.g., regional land parcel registries or thematic maps when available) and the harmonized European datasets CORINE Land Cover (CLC), developed by Copernicus Land Monitoring Service (CLMS)-European Environment Agency (EEA) [56]. These datasets are used to generate spatial masks that restrict AgroSat's remote sensing analyses to relevant agricultural zones, ensuring efficient use of computational resources and improving the accuracy of indicators such as vegetation indices and evapotranspiration.

Gridded climate data are also integrated into AgroSat via the Copernicus Climate Data Store (CDS; https://cds.climate.copernicus.eu/), an open-access repository providing global and regional climate information [57]. CDS hosts a wide range of datasets, including historical observations, satellite data, reanalysis, projections, and seasonal forecasts, all accessible through a standardized web interface. AgroSat connects to CDS through the Climate Data Store Application Program Interface (CDS-API), enabling automatic and programmatic data retrieval. Specifically, two core data products are used:

- ERA5 Reanalysis: developed by the European Centre for Medium-Range Weather Forecasts (ECMWF), ERA5 provides hourly estimates of atmospheric, land-surface, and sea-state parameters since 1950, on a regular grid [58]. It integrates historical observations with numerical weather prediction models to generate a temporally and spatially consistent dataset. ERA5 is particularly useful in regions with limited in-situ data, filling gaps in AgroSat's coverage with high-quality modeled variables such as temperature, precipitation, wind speed, soil moisture, and surface radiation [57].
- SEAS5 Seasonal Forecasts: also provided by ECMWF-System5 seasonal forecasts data, which offer predictions of key meteorological variables over extended timeframes (weeks to months), supporting AgroSat's predictive modeling features. These data are used to estimate potential impacts of upcoming weather patterns on crop growth and yield, allowing users to make anticipatory agronomic decisions [59].

2.1.4. Data cleaning: cloud and shadow detection

AgroSat integrates the AgroShadow tool into the processing workflow to enhance the accuracy of cloud shadow detection in Sentinel-2 imagery for precision agriculture applications [60]. AgroShadow is a threshold-based algorithm that leverages the Short-Wave Infrared (SWIR, Sentinel-2 band B12) band and the Normalized Difference Vegetation Index (NDVI, Sentinel-2 bands B8 and B4) from Sentinel-2 data to identify cloud shadows. Unlike other tools such as Sen2Cor and MAJA, AgroShadow does not require information on cloud location or solar geometry, simplifying its implementation. In comparative evaluations across various Italian agricultural scenarios, AgroShadow demonstrated superior performance, achieving the lowest error rates and highest accuracy and F-scores among the tested methods [60].

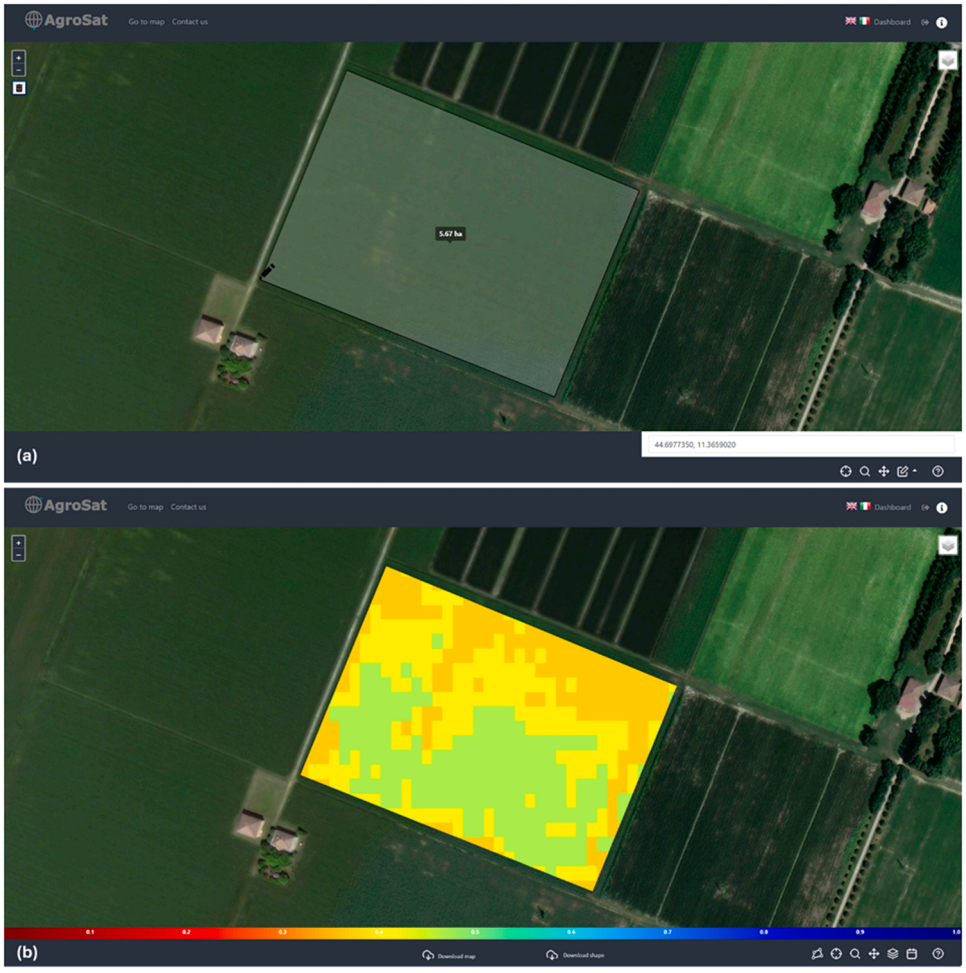


Fig. 1. User-defined production unit (PU) (a) delineated using AgroSat drawing tool and the corresponding water status map (b) with water content (cm^3/cm^3) ranging from 0 (red) to 1 (blue).

2.1.5. *Soil moisture*

For surface soil moisture (SSM) detection, AgroSat integrates the nonlinear Optical Trapezoid Model (OPTRAM) [61], specifically calibrated for use with Sentinel-2 imagery [62]. OPTRAM combines information from NDVI and SWIR spectral bands to derive SSM estimates with high spatial (10 m) and temporal resolution. This approach enables the detection of intra-field water stress dynamics and can support precision and timely irrigation interventions as well as stress monitoring (Fig. 1, b). The method has proven effective across different crop types and agro-climatic regions, providing a robust operational tool for large-scale agricultural applications [62].

2.1.6. *Potential yield map*

The potential yield map is generated using satellite-based NDVI data and known correlations between NDVI and biomass/yield from literature and internal datasets [63,64,70,71]. It expresses, on a pixel basis, the estimated productive potential of the field under optimal growing conditions. The output is presented as a percentage index (0–100 %), where 100 % represents the highest potential yield within the field. This allows users to quickly identify more and less productive zones.

2.1.7. *Precision nitrogen fertilization*

To optimize N applications by allocating inputs according to spatial field variability, AgroSat generates VRT fertilization maps based on NDVI and crop yield data. NDVI maps are derived from Sentinel-2 observations, offering high spatial and temporal resolution insights into crop vigor. Historical yield data are acquired by RTK-GPS, the real-time kinematic global positioning system. The RTK-GPS sensors, mounted on harvesting machinery, provide georeferenced yield measurements with centimeter-level accuracy (1–3 cm), allowing for precise spatial referencing [65,72,73]. As recommended by Vega et al. [67], these data are pre-processed using a cleaning protocol to remove implausible values ($<$ 0.1 t/ha), data near field boundaries (within 10 m), and statistical outliers (values outside mean $\pm$3 SD). Spatial outliers are further refined using local indicators of spatial association (LISA) such as Moran's I value [74,75]. The cleaned yield shapefiles (containing latitude, longitude, and moisture data) can then be uploaded into AgroSat to define PUs and match NDVI data.

AgroSat allows for the configuration of single or dual source output VRT N fertilization maps. A single-source map, based on NDVI values, is recommended when recent biomass conditions are the primary consideration, while a dual-source map, integrating NDVI and yield data, is recommended for more accurate long-term field performance estimation. Regarding dual-source mode, a score-based data integration method is applied by the CombSum algorithm [76–78]. This technique normalizes data layers to a common scale and sums their scores to derive a unified spatial variability index, which is used to delineate management zones (MZs) [66] and drive site-specific fertilization.

After selecting the data source, users can define the total N rate (e.g., 80 units of urea at 46 % N) and choose between two fertilization strategies: compensation and differentiation. In the compensation mode, the model assumes that the observed spatial variability within the field is primarily due to N deficiency. It applies an inverted NDVI logic, where lower NDVI indicates higher N demand. Relative weights for each pixel

are calculated as:

$$W_i = NDVI_{\max} - NDVI_i + \varepsilon \quad (1)$$

where W_i is the weight assigned to pixel i, $NDVI_i$ is the NDVI value of pixel i, $NDVI_{max}$ is the maximum NDVI value within the defined field, and ε is an adjustment constant used to avoid zero weights and ensure numerical stability.

Thereafter, weights are normalized so that the sum over the field equals one. The final N dose for each pixel is calculated as follows:

$$N_i = N_{Tot} \times W_i \quad (2)$$

Where N_i is the dose of N assigned for pixel $_i$ and N_{Tot} is the total input N defined by the user.

This ensures that the total N input defined by the user is maintained but spatially redistributed according to crop vigor.

In the differentiation mode, the model introduces a non-linear allocation logic that integrates crop performance potential to prevent unnecessary N application in underperforming areas. This approach is designed to reduce N use, typically by up to 20 %, compared to the user-specified total dose. It de-emphasizes low NDVI areas if they are likely affected by abiotic stresses (e.g., waterlogging, poor soil), rather than N deficiency, thus avoiding waste and minimizing environmental risk. The maximum redistribution observed under differentiation mode is usually around 20 %, whereas compensation mode reveals in lower adjustment amplitudes. The magnitude of redistribution is therefore field-specific and depends on the selected management strategy.

Regarding the output map resolution, users can choose between high-resolution (HR) option, which preserves the original Sentinel-2 single pixel variability, and a zonal resolution (2-class, 3-class, 4-class) that groups the field into MZs with discrete N rates (low, medium, high), suitable for variable-rate machinery. Moreover, users can modify the spatial resolution by setting the tractor implement width, enabling resampling of the output to match field operations. All maps can be queried via the interactive interface and exported in various formats, including image, shapefile (for GIS), and ISO-XML (ISOBUS standard) for compatibility with VRT machinery.

2.1.8. Meteorological data

Weather data are integrated from regional agrometeorological networks, including public Open Data platforms and private stations when available. AgroSat automatically links each PU with the closest available station (within a 10 km radius) and enables visualization of observed variables such as air temperature, relative humidity, precipitation, solar radiation, wind speed and direction, and leaf wetness. Users can select the time window of interest and download both graphical and raw formats (JPEG or CSV) for further analysis. Additionally, the system includes a weather forecast function providing 5-day hourly predictions for key parameters, enhancing decision-making capabilities for time-sensitive agricultural operations.

2.1.9. Crop phenology modeling

AgroSat integrates plant phenology modeling for barley, rapeseed, sunflower, corn, rice, soybean, and chickpea, based on growing degree days (GDD) to estimate crop development based on temperature [68,79,80]. For soft and durum wheat, the main phenological phases (emergence, booting, earing, flowering, complete maturation) are calculated by the Delphi mechanistic model [81,82] (Fig. 2). The Delphi system has an operational tool that stores and updates input meteorological, soil, and phenological data to simulate crop phenological development, biomass growth, and yield data, integrating a geographical information system for output visualization and spatialization.

2.1.10. Water balance

As for crop phenology, AgroSat calculates water balance based on GDD [68,79,80]. Crop evapotranspiration (ET_c) is calculated following the standard crop coefficient approach as $ET_c = ET_0 \times Kc$, where ET_0 is the reference evapotranspiration and K_c is the crop coefficient [69]. Daily is calculated by LSA-SAF data and K_c provided by FAO, corresponding to the crop phenological stage. The estimated evapotranspiration in AgroSat refers exclusively to the rainfall, without including other water-input sources such as irrigation systems. Potential evapotranspiration (PET) is estimated from radiation balance and air-drying power and is divided into potential transpiration and potential evaporation. Penman-Monteith or Priestley-Taylor methods [69] are applied for estimating PET, depending upon the availability of weather data. The

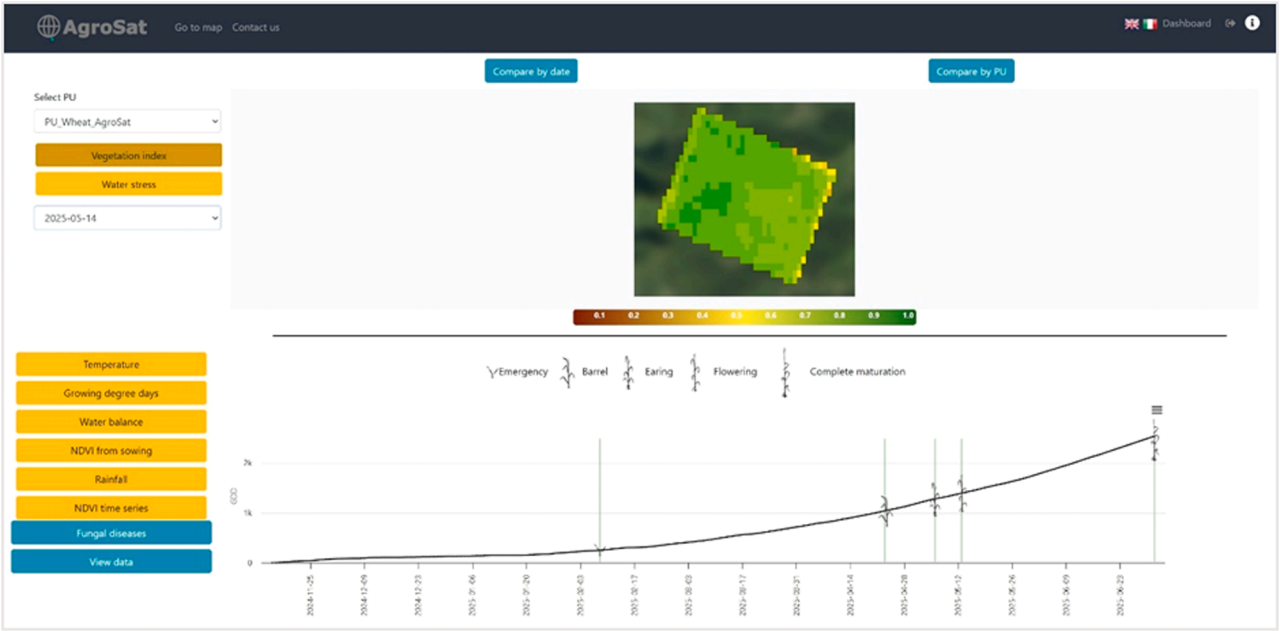


Fig. 2. Analytical tools available on AgroSat-Studio environment for a user-defined PU of a wheat field. The upper part of the figure shows the NDVI map at flowering stage (May) and the lower part shows the phenological growth estimated based on the growing degree days (GDD). The NDVI color scale ranges from 0 (brown) to 1 (dark green).

Penman-Monteith method calculates PET based on radiation balance, air-drying components, and leaf area index, while the Priestley-Taylor method is based only on radiation balance [83].

2.1.11. Seasonal forecast-based yield and quality prediction

AgroSat incorporates a dedicated tool for forecasting durum wheat yield and grain protein content based on seasonal climate forecasts, as developed and tested by Dainelli et al. [59]. The model leverages the unbiased ECMWF SEAS5 seasonal forecasting system and integrates gridded agro-meteorological indicators to drive the Delphi model [81, 82] for yield and quality prediction. The system is calibrated using harmonized historical data from Italian provinces and validated through cross-year testing. Forecasts are updated monthly, providing users with probabilistic projections for both productivity and quality attributes several months ahead of harvest.

2.2. Spatial data infrastructure, system architecture, and user interfaces

Spatial Data Infrastructure (SDI) is an essential prerequisite for a well-structured precision farming platform. It enables efficient production, visualization, and dissemination of multi-source spatial information by integrating systems, networks, standards, and institutional frameworks [84–86]. A process-based SDI conceptual model [85] was adopted instead of the product-based one [87] to ensure more active and flexible geo-data processing, efficient visualization, and user-specific outputs. AgroSat is built on a Service-Oriented Architecture (SOA) [88] and implements Open Geospatial Consortium (OGC) standards [89, 90] to ensure interoperability, scalability, and component independence. These standards are applied across the platform, from the ISO TC/211 UML-based data model to the PostgreSQL/PostGIS GIS engine, which supports OGC Simple Feature Specifications for SQL [91] and enables efficient geospatial data management.

The architecture of AgroSat comprises three core layers. First, the Providers Layer handles data acquisition from multiple sources, supporting OGC-compliant raster formats (e.g., NetCDF for Sentinel-2 and ERA5; HDF for LSA-SAF and EUMETSAT), vector formats (WKT and GeoJSON), and JSON for meteorological data [92]. Second, the Framework Layer functions as the operational core and relies on two PostgreSQL databases operating in parallel: CoreDB, for storing analytical procedures and results together with ancillary datasets and vector and raster data; SentinelDB, for storing Sentinel satellite bands and precomputed NDVI and RGB composite together with the analytical procedures required to generate derived products on-the-fly. Third, the Client Layer enables user interaction through the AgroSat and AgroSat-Studio web applications, built with PHP/Ajax and RESTful services to ensure interoperability and third-party access. These REST services, selected for their lightweight and client-friendly design, support efficient data access, visualization, and service delivery [93,94]. Table 2 describes relationships between Agrosat products to their inputs and processing steps.

To further improve the response time of computational procedures, the two databases are deployed on separate dedicated Linux servers, each providing the DBMS with 20 CPU cores and 80 GB of RAM. The execution of calculation procedures is orchestrated by a third dedicated Linux server (AgroSatWS), which manages J2EE web services responsible for brokering procedure calls, supporting multithreading, and handling asynchronous processes, including both on-the-fly and scheduled tasks. The AgroSatWS server provides up to 16 CPU cores and 60 GB of RAM. Thanks to these optimizations, AgroSat performs many calculations on-the-fly, such as modelling soil water balance for a 120-hectare PU in less than 4 s from input retrieval to output generation. The measured time includes data retrieval, processing, and output composition, but excludes data transfer to the client, which depends on user-side network connectivity and congestion conditions.

The AgroSat interface is designed for maximum usability and adaptability, serving both advanced users and less experienced farmers.

Table 2
Products table map of AgroSat.

Agrosat Product	Input	Processing steps	Spatial/ Temporal Resolution	Export formats
Precalculated Products				
NDVI	Sentinel-2A/B/C (B4,B8)	Calculated and stored from downloaded Sentinel bands	10 m spatial resolution, 20-day temporal resolution	GeoTIFF, GeoJSON, PNG
RGB Composite	Sentinel-2A/B/C (B2,B3,B4)	Calculated and stored from downloaded Sentinel bands	10 m spatial resolution, 20-day temporal resolution	GeoJSON, PNG
On-the-fly Products				
Production Yield	Precalculated NDVI	Calculated by PL/pgSQL, procedure released by APIs and PHP	10 m spatial resolution, 20-day temporal resolution	GeoTIFF, GeoJSON, PNG
Soil Mosture	Precalculated NDVI, stored Sentinel-2A/B/C (B12)	Calculated by PL/pgSQL procedure released by APIs and PHP	10 m spatial resolution, 20-day temporal resolution	GeoTIFF, GeoJSON, PNG
Prescription Map	Precalculated NDVI	Calculated by PL/pgSQL procedure, released by APIs and PHP procedures	10 m spatial resolution, 20-day temporal resolution	GeoTIFF, GeoJSON, PNG
Phenology	ERA5, ETP, Crop ancillary data	Calculated by PL/pgSQL procedure, released by PHP procedure	0.25° spatial resolution, daily temporal resolution	CSV, JPEG
Water Balance	ETP, ERA5	Calculated by PL/pgSQL procedure, released by PHP procedure	0.25° spatial resolution, daily temporal resolution	CSV, JPEG

Upon registration, users access a dashboard offering a wide array of services, including field boundary management, nitrogen prescription maps, weather visualization, climate trends, and model outputs. The platform is fully responsive across devices, from desktop to mobile, including real-time GPS-guided navigation of prescription maps. The graphical user interface (GUI) integrates input from farmers, agronomists, and researchers, and emphasizes clarity, intuitive navigation, and rapid access to decision-support tools. User-provided data (e.g., field records, phenological stages, and management actions) directly inform output customization and model calibration.

Overall, AgroSat integrates a back-end optimized SDI with a user-oriented front-end, ensuring accessible and data-driven decision-making at all levels of expertise. Fig. 3 illustrates the spatial data infrastructure and system architecture of AgroSat, highlighting the data flow across the above-mentioned core functional layers.

AgroSat-based VRT nitrogen fertilization: a case study

2.3. Case study objective and scope

This case study aimed at assessing the potential of AgroSat variable-rate nitrogen fertilization in standardizing intra-field yield and crop vigor. The study area consisted of eight winter wheat (*Triticum aestivum* L.) fields located in Parma, Italy (36 m a.s.l., 44.82° N, 10.37° E), covering a total area of approximately 22.7 ha, namely Field_1 (3.76 ha), Field_2 (2.67 ha), Field_3 (2.48 ha), Field_4 (3.07 ha), Field_5 (1.00 ha), Field_6 (3.22 ha), Field_7 (3.22 ha), and Field_8 (3.26 ha) (Fig. 4). Fields

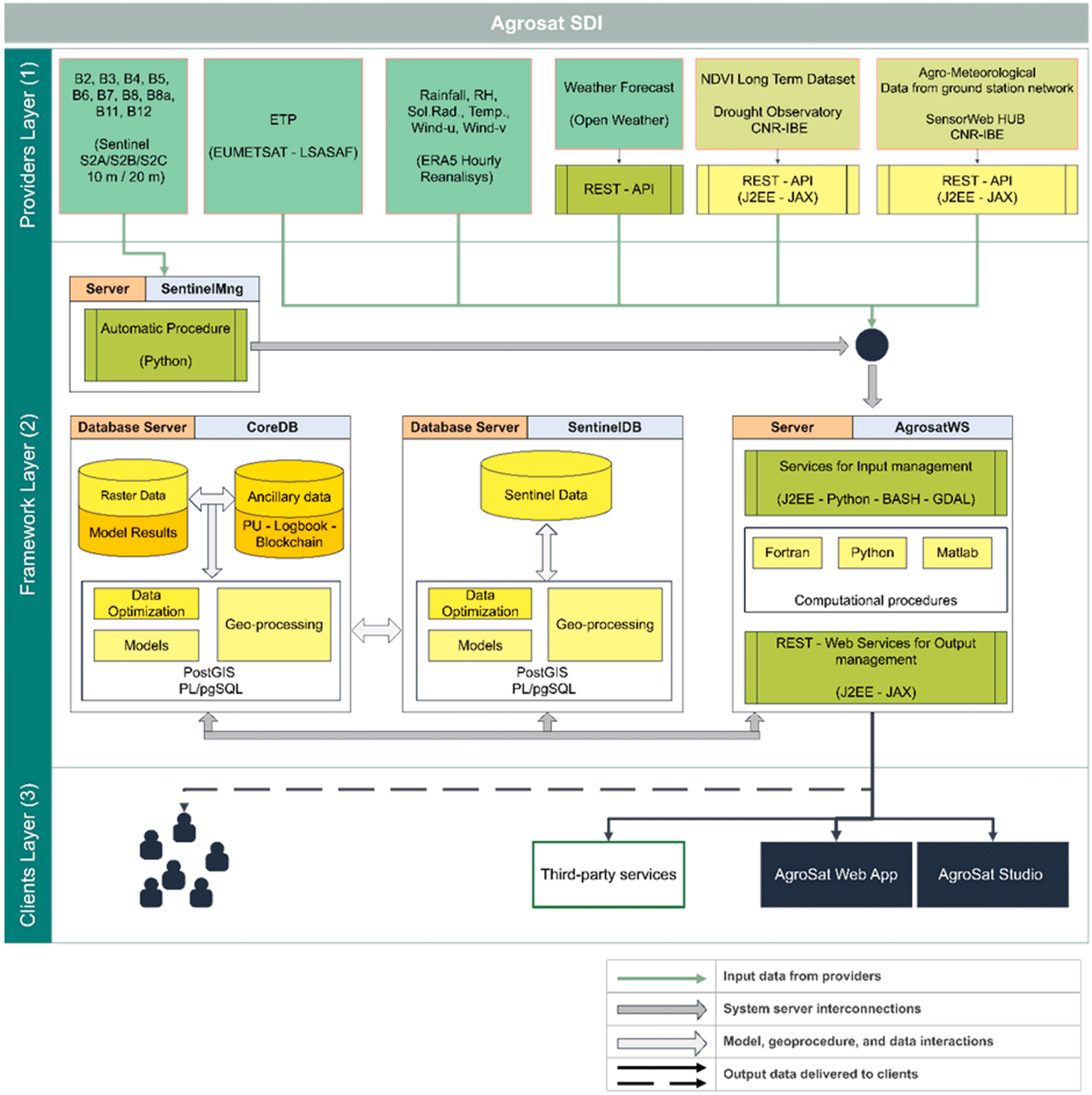


Fig. 3. Conceptual diagram of the AgroSat Spatial Data Infrastructure (SDI) and system architecture. The system is structured into three core layers: (1) Providers Layer, responsible for supplying multi-source input data to the system; (2) Framework Layer, which manages server interconnections, geoprocessing models, and data interactions; and (3) Client Layer, where processed outputs are delivered to end-users.

were characterized by heterogeneous yield potential and variable soil conditions. The AgroSat platform was used to delineate MZs and generate field-specific N fertilization maps by integrating user field data, satellite imagery, and geostatistical analyses. Spatially explicit yield distribution and NDVI time-series variability were assessed to characterize within-field performance and evaluate AgroSat VRT outcomes across fields.

2.4. AgroSat VRT strategy

Winter wheat was sown on 4 January and harvested on 8 July using John Deere T670i, a combine harvester equipped with a yield mapping system. During the growing season, the experimental fields underwent two N fertilization events, VRN_03 and VRN_05, applied on 25 March (fertilizer containing 26 % N) and 3 May (fertilizer containing 46 % N), respectively. The fertilization was performed by a Khun VRT fertilizer spreader connected to the Topcon system console through the ISOBUS standard. The recommended rates were determined using a N balance model and served both as the amount for the control treatment and as the input for generating AgroSat N fertilization maps. The experimental design included three-class VRT maps for Field_1, 2, 3, 4 and 6, corresponding to three treatment zones of N (LN, low nitrogen; MN, medium nitrogen; HN, high nitrogen), and two-class VRT maps for Fields_7 and 8 (LN and HN). Field_5 was designated as the control field and managed under traditional, uniform-rate N fertilization (Fig. 4). The N treatment zones (LN, MN, HM) were distributed within each field following the spatial variability patterns driven by NDVI values. To select the proper NDVI acquisition date, Sentinel-2 images acquired between sowing and the first treatment period (February–March) were considered for the first N fertilization (VRN_03), while images acquired between the first

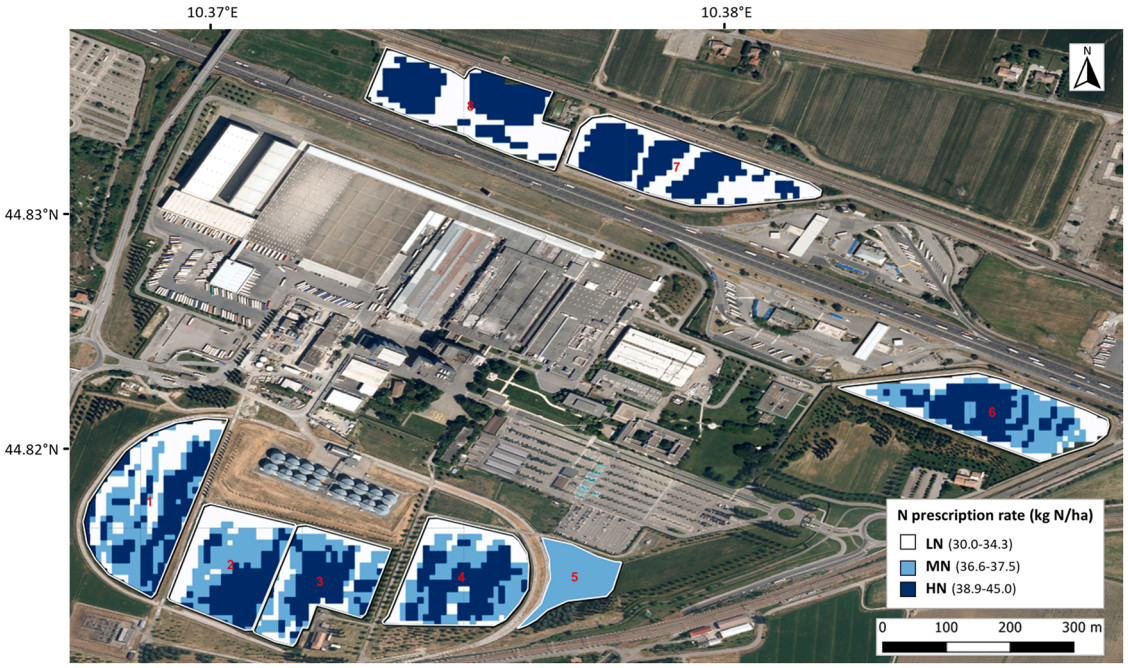


Fig. 4. Study area and variable-rate nitrogen fertilization map at VRN_03 (March treatment) of the eight experimental fields. Field_5 represents the control field with uniform-rate N fertilization. LN = Low Nitrogen rate; MN = Medium Nitrogen rate; HN = High Nitrogen rate. Color classes represent relative management zones (LN/MN/HN) derived from intra-field variability analysis; corresponding numeric ranges are reported in the legend.

and second treatment (March–May) were analyzed for the second N fertilization (VRN_05). NDVI values retrieved on 15 March and 27 April were then selected as a reference indicator of spatial crop N requirement at VRN_03 and VRN_05, respectively. The N doses recommended for each single field and the N content (%) present in the commercial fertilizer were entered into the AgroSat platform to generate single-source VRT prescription maps based on the compensation mode (see Section 2.1.7). The generated VRT maps followed a color gradient ranging from white (lower N application rates) to dark blue (higher N application rates) (Fig. 4).

2.5. Analytical approach

The effectiveness of AgroSat variable-rate N fertilization was evaluated in terms of intra-field grain yield variability and crop vigor (NDVI). To generate georeferenced yield maps, yield data were recorded by the combine harvester on-board computer with a spatial sampling interval of approximately 1 m along the harvesting direction. Thereafter, yield



Fig. 5. Wheat grain yield maps (10 m resolution) after the cleaning process of the eight experimental fields. Field_5 represents the control field with uniform-rate N fertilization. Map colors represent absolute grain yield (t/ha).

data were pre-processed to remove spatial outliers and measurement noise prior to the delineation of management zones. The cleaning procedure followed a multi-step approach including: (i) removal of edge effects and non-harvested areas; (ii) filtering of extreme values using a ±3 standard deviation threshold; and (iii) spatial outlier detection using LISA based on Moran's I [74,75]. Spatial autocorrelation analysis was performed using a first-order queen contiguity spatial weights matrix and a significance threshold of p-value < 0.05. Observations identified as significant spatial outliers (high–low or low–high clusters) were removed prior to interpolation and clustering. Overall, approximately 25 % of the original yield observations were excluded during the cleaning process, consistent with previous applications of the same workflow in comparable yield-monitor datasets [63]. The same spatial framework was applied to support the delineation of homogeneous management zones, ensuring consistency between yield data cleaning and spatial clustering procedures and improving the robustness of zone-based agronomic analysis.

The resulting clean yield maps, made available in AgroSat as shapefiles, are shown in Fig. 5. For each field, grain yield data were aggregated by VRT nitrogen level zone and analyzed independently to focus on within-field variability rather than comparisons between fields. VRT management performance was additionally evaluated using nitrogen use efficiency (NUE), estimated as PFP, a nitrogen partial factor productivity obtained from the ratio between kg of harvested grain and kg of N input. Differences in mean yield and NUE across the different VRT fertilization zones were assessed using analysis of variance (ANOVA), followed by Tukey's honestly significant difference (HSD) post-hoc test for pairwise comparisons (p-value < 0.05). Yield uniformity was further characterized using descriptive statistics, including standard deviation (SD), variance, and 95 % confidence interval (CI). Moreover, NDVI time-series variance was evaluated for each field to assess the intra-field variability of crop status before and after fertilization treatments. The relationship between yield and NDVI after the VRT fertilization on 13 June was also evaluated, with the coefficient of determination (R-sq) and p-value reported to assess statistical significance.

3. Results

The effects of AgroSat-based VRT nitrogen fertilization on grain yield and NUE are reported in Table 3. For each field, results are shown per N application timing to compare yield variability between VRT MZs after the two fertilization events, VRN_03 and VRN_05. Overall, results demonstrated that AgroSat-based VRT treatments reduced grain yield variability in most experimental fields (Fig. 5). Statistical analysis showed no significant differences in mean yield at each fertilization time, reflecting the uniformity of yield distribution across N treatment zones (Table 3). Exceptionally, Field_7 exhibited significant differences at VRN _05, with a higher mean yield of 5.38 t/ha observed with LN, compared to 5.08 t/ha with HN treatment. Average yield was the highest in Field_1 with 7.62 t/ha, followed by Field_2 and 3 with almost 6.88 t/ha, while Field_6 and Field_7 showed the lowest values of 4.93 and 5.23 t/ha, respectively. Yield variance was remarkably reduced across the VRT-managed zones, associated with relatively narrow 95 % CI of mean yield. The lowest range of yield variance (0.01–0.12) was observed for Field_1,2,3,4, and 6, while slightly higher variance occurred in Field_7 (0.09–0.19). Field_8 revealed the highest yield variance, ranging between 0.24 and 0.34, which is higher than that of Field_5. When considering the overall within-field variance, a clear reduction in yield variability was detected across most VRT-managed fields compared with the uniform-rate control field. In particular, within-field yield variance in VRT fields ranged from 0.02 to 0.16, substantially lower than Field_5 variance (0.24), indicating a more homogeneous spatial distribution of production. The only exception was Field_8, where known soil constraints influenced yield variability independently of nitrogen management. This pattern suggests that VRT primarily acted through a spatial compensation mechanism, reducing yield dispersion and stabilizing crop performance across heterogeneous field conditions rather than generating significant differences in mean yield among zones. Regarding the NUE, values increased as the VRT N level decreased with significant differences detected at VRN_05 in Field_1, 7, and 8, and at both fertilization timings in Field_4. NUE increased in parallel with yield under VRT fertilization, with the highest value of 65.96 kg/kg recorded in Field_1 (highest yield of 7.62 t/ha), and the lowest value of 41.36 kg/kg recorded in Field_6 (lowest yield of 4.93 t/ha). It is important to note that the NUE indicator used in this study, calculated as PFP, provides a practical evaluation due to its simplicity and direct relation to fertilizer management. However, it primarily reflects agronomic performance and does not capture the physiological efficiency of nitrogen uptake or potential environmental nitrogen losses. Therefore, the results should be interpreted as improvements in agronomic nitrogen efficiency under specific experimental conditions.

In line with yield uniformity, time-series NDVI maintained a low within-field variability under AgroSat-based VRT management, reflecting more homogeneous spatial crop vigor across the N fertilization zones (Fig. 6). Noticeable stabilizing effect of the fertilization treatments was mainly observed in Field_1, 3, 4, and 6, where reduced NDVI variance remained under 0.01. NDVI low variability is in accordance with yield uniformity in these fields. Although Field_2 and 8 maintained low variance values throughout the season, they did not show a marked effect of stabilization following the VRN_05 treatment. Field_7 demonstrated a slightly lower reduction in NDVI variability, reaching 0.02 on 01 June. This result, which is in association with increased yield variability in Field_7, is likely due to soil heterogeneity related to highway construction work that was taking place near the area. As expected, Field_5, which served as the control with uniform-rate fertilization, exhibited a fluctuating time-series trend with higher within-field NDVI variability, reaching 0.03, compared to VRT-managed fields.

Overall, results revealed a satisfactory performance, showing increased intra-field uniformity in grain yield in almost all fields, mirroring the stabilization in crop vigor under AgroSat-based VRT fertilization. This uniformity was also explained by the strong relationship between yield and post-VRT NDVI, which revealed a significant positive association (R-sq = 0.77, p-value < 0.001) (Fig. 7), confirming the research evidence about the consistency between vegetation indices and yield responses [63,95,96]. These findings support the interpretation that the main agronomic benefit of the AgroSat-driven VRT strategy lies in reducing intra-field yield variability and enhancing spatial yield stability, which is a key objective of precision nutrient management. Given that NUE was estimated as PFP, the study therefore validated the effectiveness of AgroSat-based VRT fertilization in improving nitrogen fertilizer use efficiency in terms of yield response, whereas physiological N utilization and associated environmental emission pathways remain beyond the scope of this study. Accordingly, further in-depth investigations are required to assess plant-level N dynamics and environmental impacts.

As shown by the case study, AgroSat is among the few freely accessible web platforms providing both fertilization and yield maps simultaneously. Previous field-scale studies leveraging satellite imagery have generally been limited to delineating uniform MZs, either generic [45] or crop-focused fertilization strategies in vineyard [97], rice [98] or wheat [99–101]. However, these studies did not provide product maps directly usable by end-users. For yield maps, comparable DSS products exist; for example, in sugarcane [102], satellite data were used to monitor harvesting progress and provide early-warning alerts in case of anomalies, although this involved only two pilot farmer cooperatives. Other studies focused on larger areas, such as for maize [103], or on comparing alternative methods for site-specific wheat yield mapping [104], but these efforts have largely remained at the research stage. In comparison, AgroSat represents a freely available DSS tool specifically designed for farmers and technicians, enabling direct field-level management and delivering zonal prescription maps and other actionable

Table 3
Effect of variable rate nitrogen (VRT N) fertilization on wheat grain yield and nitrogen use efficiency (NUE). Treatments: LN, low nitrogen; MN, medium nitrogen; HN, high nitrogen. Yield statistics for each VRT zone includes mean value, standard deviation (SD), variance, and 95 % confidence interval (CI). Means with different letters are significantly different (p-value < 0.05).

Field (Area, ha)	Fertilization Time	VRT Zone	n Pixel	VRT N (kg/ha)	Pre-VRT NDVI	NUE (PFP; kg Yield/ kg N)	Mean Yield (t/ ha)	SD Yield (t/ha)	95 % CI yield	Variance Yield (t/ha)	Within-field Variance Yield (t/ ha)
Field_1 (3.76)	VRN_03	HN	102	40.57	0.38	58.25 [a]	7.52 [a]	0.29	(7.38 – 8.08)	0.08	0.1
		MN	135	37.22	0.41	66.43 [a]	7.68 [a]	0.31	(7.63 – 7.89)	0.09	
		LN	130	33.65	0.48	74.37 [a]	7.66 [a]	0.33	(7.52 – 8.14)	0.11	
	VRN_05	HN	90	105.87	0.88	52.38 [b]	7.55 [a]	0.34	(7.45 – 8.29)	0.12	
		MN	146	82.45	0.92	63.94 [b]	7.64 [a]	0.29	(7.58 – 7.84)	0.08	
		LN	131	60.49	0.92	80.36 [a]	7.67 [a]	0.33	(7.48 – 8.15)	0.11	
Field_2 (2.67)	VRN_03	HN	88	44.33	0.29	49.19 [a]	6.77 [a]	0.19	(6.70 – 7.24)	0.04	0.04
		MN	86	37.54	0.36	59.96 [a]	6.90 [a]	0.16	(6.87 – 7.04)	0.03	
		LN	88	30.01	0.47	68.46 [a]	6.97 [a]	0.20	(6.78 – 7.34)	0.04	
	VRN_05	HN	88	101.93	0.90	47.73 [a]	6.81 [a]	0.20	(6.72 – 7.22)	0.04	
		MN	87	82.69	0.91	58.20 [a]	6.93 [a]	0.18	(6.90 – 7.08)	0.03	
		LN	87	62.94	0.91	71.80 [a]	6.90 [a]	0.20	(6.75 – 7.26)	0.04	
Field_3 (2.48)	VRN_03	HN	75	38.86	0.29	52.73 [a]	6.95 [a]	0.15	(6.81 – 7.20)	0.02	0.03
		MN	81	36.67	0.33	57.92 [a]	6.85 [a]	0.16	(6.81 – 6.97)	0.03	
		LN	85	34.30	0.38	63.27 [a]	6.87 [a]	0.16	(6.83 – 7.20)	0.03	
	VRN_05	HN	81	98.55	0.92	50.74 [a]	6.92 [a]	0.15	(6.81 – 7.22)	0.02	
		MN	80	83.47	0.92	56.96 [a]	6.83 [a]	0.16	(6.79 – 6.95)	0.03	
		LN	80	68.37	0.93	66.97 [a]	6.91 [a]	0.16	(6.84 – 7.26)	0.03	
Field_4 (3.07)	VRN_03	HN	99	39.48	0.27	50.70 [b]	6.48 [a]	0.12	(6.41 – 6.67)	0.01	0.02
		MN	105	36.60	0.29	54.70 [ab]	6.54 [a]	0.14	(6.52 – 6.63)	0.02	
		LN	102	33.06	0.36	59.14 [a]	6.52 [a]	0.11	(6.45 – 6.73)	0.01	
	VRN_05	HN	103	94.58	0.92	48.90 [b]	6.46 [a]	0.13	(6.39 – 6.71)	0.02	
		MN	102	82.84	0.92	54.51 [ab]	6.52 [a]	0.12	(6.49 – 6.60)	0.01	
		LN	101	72.46	0.89	61.37 [a]	6.56 [a]	0.11	(6.48 – 6.79)	0.01	
Field_5 (1.00)	VRN_03	MN	97	36.92	0.30	55.69 [a]	6.62 [a]	0.49	(6.52 – 6.94)	0.24	0.24
	VRN_05	MN	97	81.88	0.86	55.69 [a]	6.62 [a]	0.49	(6.52 – 6.94)	0.24	
Field_6 (3.22)	VRN_03	HN	107	41.67	0.32	37.13 [a]	4.88 [a]	0.24	(4.88 – 5.30)	0.06	0.06
		MN	109	37.41	0.37	41.30 [a]	4.94 [a]	0.25	(4.88 – 5.04)	0.06	
		LN	89	32.46	0.42	45.30 [a]	4.95 [a]	0.22	(4.68 – 5.15)	0.05	
	VRN_05	HN	114	95.47	0.89	36.16 [a]	4.88 [a]	0.24	(4.80 – 5.22)	0.06	
		MN	103	82.68	0.88	41.14 [a]	4.91 [a]	0.24	(4.85 – 5.02)	0.06	
		LN	88	70.95	0.88	47.12 [a]	4.99 [a]	0.23	(4.80 – 5.28)	0.05	
Field_7 (3.22)	VRN_03	HN	155	44.99	0.17	35.98 [a]	5.07 [a]	0.29	(5.09 – 5.53)	0.08	0.16
		LN	159	30.27	0.29	58.27 [a]	5.39 [a]	0.43	(5.13 – 5.56)	0.18	
	VRN_05	HN	157	111.96	0.69	33.29 [b]	5.08 [b]	0.29	(5.34 – 5.82)	0.09	

(*continued on next page*)

Table 3 (*continued*)

Field (Area, ha)	Fertilization Time	VRT Zone	n Pixel	VRT N (kg/ha)	Pre-VRT NDVI	NUE (PFP; kg Yield/ kg N)	Mean Yield (t/ ha)	SD Yield (t/ha)	95 % CI yield	Variance Yield (t/ha)	Within-field Variance Yield (t/ ha)
		LN	157	57.05	0.73	61.24 [a]	5.38 [a]	0.44	(4.86 – 5.31)	0.19	
Field_8 (3.26)	VRN_03	HN	153	43.32	0.36	43.79 [a]	5.94 [a]	0.58	(5.75 – 6.47)	0.34	0.30
		LN	163	31.62	0.47	62.32 [a]	6.13 [a]	0.49	(5.96 – 6.66)	0.24	
	VRN_05	HN	157	108.10	0.86	39.89 [b]	5.89 [a]	0.56	(5.93 – 6.81)	0.31	
		LN	159	59.72	0.89	66.64 [a]	6.18 [a]	0.49	(5.71 – 6.56)	0.24	

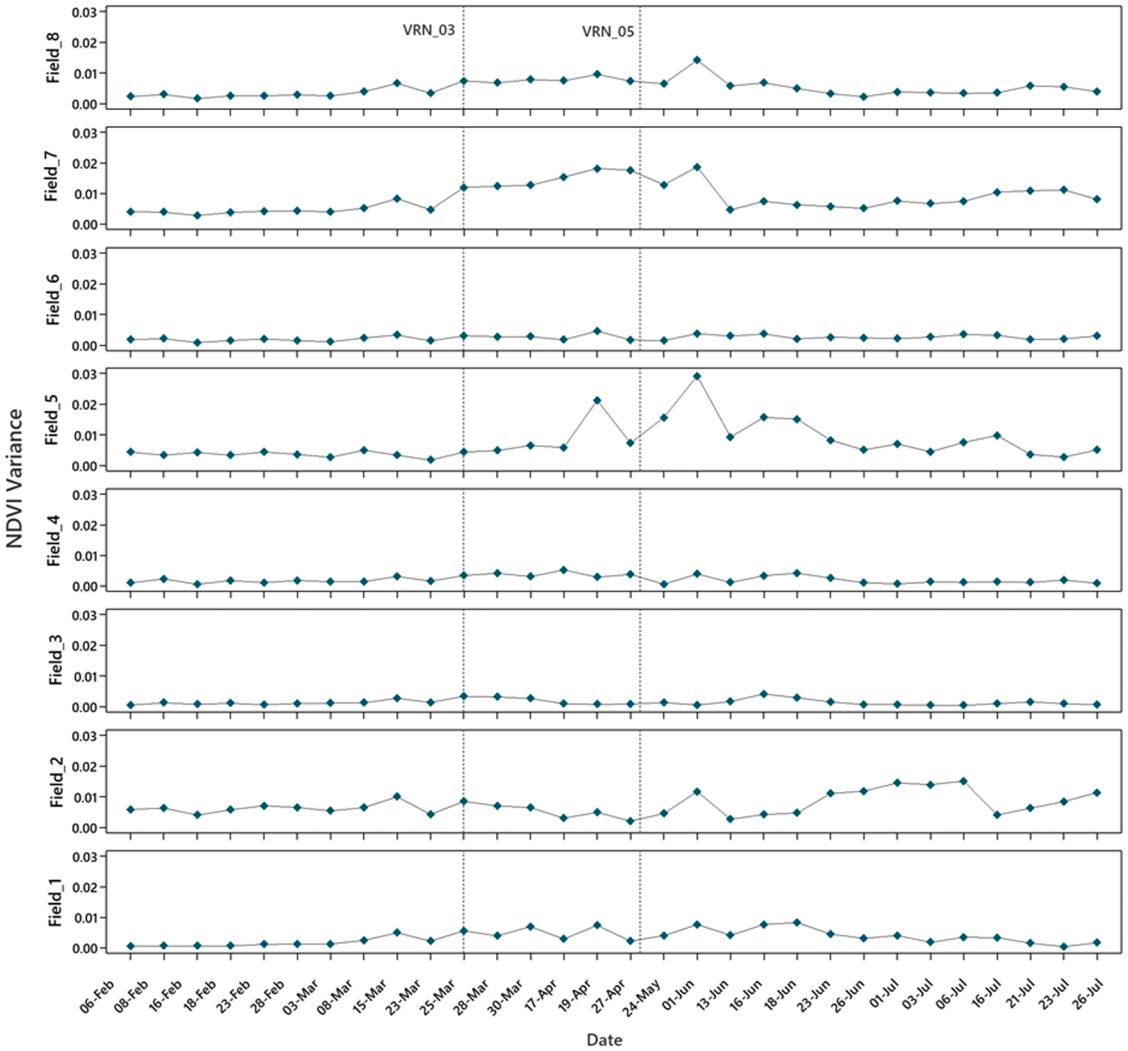


Fig. 6. Time-series NDVI variance of the eight experimental fields from emergence to harvest (data from Sentinel-2 satellites). The fertilization events occurred on 25 March (VRN_03) and 03 May (VRN_05).

products to end-users.

Despite the promising results obtained in this study, some limitations should be acknowledged. The present analysis focuses on a specific crop and on a limited number of experimental fields within a defined agro-environmental context. Consequently, the quantitative outcomes reported here should be interpreted primarily at field and regional scale. In particular, the current results demonstrate AgroSat's effectiveness in regulating spatial yield uniformity and crop vigor within the same growing season, while the temporal stability of the zoning strategy may also be influenced by inter-annual climate variability as well as longer-term changes in soil fertility, which should be addressed through multi-season validation. However, AgroSat has been designed as a crop-independent and scalable decision-support framework. The platform integrates multi-source satellite, climatic and field-level data through a flexible and transferable architecture that can be adapted to different cropping systems and environmental conditions. Beyond winter wheat applications presented here, AgroSat is currently implemented in both operational farm contexts and research projects across different cereal

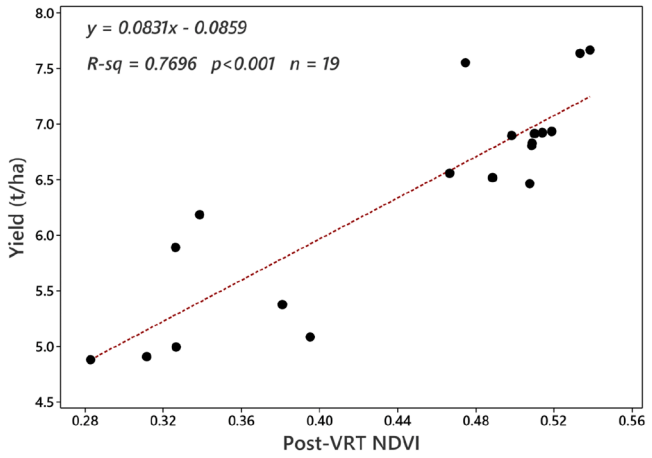


Fig. 7. Relationship between average grain yield (t/ha) and NDVI after the variable rate technology (VRT) fertilization (13 June). Each data point represents the average value of all pixels within the respective VRT zone, with the number of pixels per zone reported in Table 3. Regression equation, coefficient of determination (R-sq), significance level (*p*-value), and sample size (n) are reported.

systems and additional agricultural sectors, including viticulture. Although extended multi-regional and long-term validation is still ongoing, the results obtained so far support the robustness and transferability of the methodological approach.

4. Impact and perspectives

AgroSat, developed by CNR-IBE, is recognized by Copernicus as the sole use case platform dedicated to Italian agriculture (https://www.copernicus.eu/en/use-cases/agrosat). It represents a significant advancement in the integration and operationalization of precision agriculture tools, bridging scientific research and practical farm management. By leveraging multi-source data, including satellite imagery (Sentinel-2, MODIS), proximal observations, and gridded climate datasets, AgroSat provides a comprehensive framework for monitoring intra-field variability through site-specific, data-driven decision support. Its interoperable SDI and adherence to OGC standards ensure flexibility, scalability and seamless integration with existing geographic information systems, enabling researchers and practitioners to access and analyze data efficiently. The platform employs data fusion and mechanistic models to provide useful insights into crop performance and the design of VRT fertilization strategies, validated through extensive field experiments in arable crops. These features not only enhance the methodological rigor of agronomic research but also promote reproducibility and benchmarking of precision farming models.

The present study is intended as an overview of the AgroSat platform functioning under real operational conditions rather than as a dedicated multi-site validation experiment. The system builds on previous field trials and methodological developments and is currently used in operational contexts by a large and growing community of users across different crops and regions. The case study was therefore selected to illustrate platform behavior under heterogeneous field conditions and practical deployment scenarios. The consistency of the results obtained across the analyzed fields, together with extensive operational use, supports the robustness of the proposed approach.

From an operational perspective, AgroSat delivers actual benefits to a large community of users, including farmers, agronomists, cooperatives, consortia, and decision-makers. A co-design approach is continuously adopted for enabling end-users to access and work on the platform easily and for ensuring long-term adoption. This activity encompasses all phases of the DSS development process, from the initial user needs assessment, through the testing phase, to the final technical dissemination phase in which the platform's features are released. The user-friendly web interface of AgroSat overcomes the issues related to software installation, allowing field-level management through georeferenced production units, customizable nitrogen prescription maps, and real-time visualization of crop growth, water balance, and yield potential. Its capacity to integrate historical and forecasted data supports anticipatory management decisions, mitigating risk and enhancing resource efficiency. By providing final users with spatially detailed and temporally consistent information, AgroSat enables more sustainable management of fertilization, irrigation and crop monitoring practices. Moreover, as a free and openly accessible DSS, AgroSat broadens access to advanced digital agriculture tools, fostering adoption among small- and medium-scale farmers and enabling evidence-based precision management.

Ongoing developments of the AgroSat platform are oriented toward the integration of advanced analytical modules to further enhance predictive capabilities and operational usability. In particular, future implementations will explore the incorporation of machine learning approaches for improving multi-source data fusion, crop status assessment, and input management optimization. For example, time-series satellite data combined with machine learning models could be used to predict the optimal nitrogen application window for each management zone, enabling more precise and timely interventions. In addition, new services are under development to support soil-related indicators, including soil organic carbon (SOC) estimation and monitoring. In parallel, the integration of user-feedback and interaction mechanisms is being considered to refine decision-support outputs and better align platform functionalities with farmers' operational needs. For instance, user interface features could allow farmers to rate the effectiveness of prescription maps during the season, supporting iterative improvement of the zoning algorithm. In future releases, a dedicated data-quality panel will be implemented within the user interface to further enhance transparency and operational usability. All these developments aim to strengthen AgroSat as an adaptive and scalable digital agriculture framework capable of supporting data-driven management across diverse cropping systems and environmental conditions.

Ethics statement

Not applicable: This manuscript does not include human or animal research.

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Data availability

The input data used by AgroSat are available in publicly accessible repositories. AgroSat products and services are available directly through the platform's website (https://www.agrosat.it/). The agronomic data of the case study are available from the corresponding author upon reasonable request.

CRediT authorship contribution statement

Najwane Hamie: Writing – review & editing, Writing – original

draft, Visualization, Formal analysis, Data curation. **Riccardo Dainelli:** Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Formal analysis, Data curation. **Leandro Rocchi:** Writing – review & editing, Writing – original draft, Software, Methodology, Data curation, Conceptualization. **Alessandro Matese:** Writing – review & editing, Validation. **Salvatore Filippo Di Gennaro:** Writing – review & editing, Validation. **Mirvat Redwan:** Writing – review & editing, Data curation. **Piero Toscano:** Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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